

# 1 Main View with Tiles

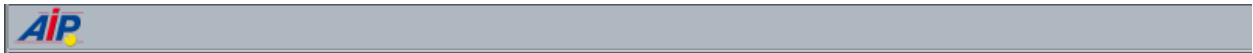
With the AIP2, the user can switch between the tile design optimized for touchscreens and the list format. By default, the tile layout is shown, which is described in the sections that follow.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

## 1.1 Main view – header and footer

### Header



The AIP logo is displayed on the top left of the screen, which may be replaced with a customer logo after configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

### Main views

You can assign a maximum of 16 workplaces or machines to the AIP2 terminal. The different workplaces are listed in the order that they were assigned to the terminal on the client. .

In the main view of the AIP2, you can use the button "< Overview" to switch to the icon view of workplaces. In the terminal configuration of the client you can specify whether you want to use the icon view. The sections that follow describe the main view and the icon view.

### Footer



The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where you can start further administration functions. This dialog closes automatically after approx. 5 seconds.

In the middle, further information is displayed: the current terminal status, AIP2 version number, date of the build, IP address of the server and the terminal number.

The current date and time are displayed to the right.

## AIP2 statuses

|  |                                                                                                                                                                                                                                                                                                                                   |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Network connection has been established.<br>The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.                                                                                                                                                                                |
|  | The terminal is sending data to the server.                                                                                                                                                                                                                                                                                       |
|  | No network connection or no connection to the server.<br>The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But you can still record certain postings. These postings are transferred to the server, once data connection has been re-established. |
|  | Data is being received.<br>The terminal reads files from the server or writes data to the server.                                                                                                                                                                                                                                 |
|  | The terminal is sending saved data records to the server.                                                                                                                                                                                                                                                                         |
|  | DEMO mode<br>The terminal is in DEMO mode, i.e. server communication is disabled.                                                                                                                                                                                                                                                 |

## 1.2 Main view with "tiles"

**List of workplaces**

| Workplace/machine | Status          | Quantity |
|-------------------|-----------------|----------|
| 33021             | PRODUCTION      | 833      |
| 40312             | OUT OF MATERIAL | 1381     |
| 406457A           |                 | 2500     |
|                   |                 | 13       |
|                   |                 | 66       |
|                   |                 | 0        |
| 79667             | PRODUCTION      | 51       |
| 79668             | PRODUCTION      | 78       |
|                   |                 | 0        |

**Workplace tiles**

| Workplace/machine | Short name | Group  | Status     |
|-------------------|------------|--------|------------|
| 33021             | WPL02      | FINISH | PRODUCTION |

**Operation tiles**

| MES order number | Article           | Quantities (target/yield/scrap) |
|------------------|-------------------|---------------------------------|
| S30002090200     | SAR-VPO011-M-9005 | 1200 / 660 / 0                  |

**Staff tiles/ Resource tiles/ Material tiles**

| Person   | Name           | Status |
|----------|----------------|--------|
| 10002630 | White, Susan   |        |
| 85741    | Meyers, Joseph |        |
| 40563    | Green, Mary    |        |
| 10002854 | Baker, Jake    |        |

Please note: The actual display can be different to the above illustration.

Subject to the configurations made, the main view with tiles consists of two or three rows of tiles. While the first two rows of tiles (workplace and operation tiles) are always displayed, it is up to the user whether or not the third row of tiles is shown (optional display). In the configuration of workplaces you can configure for each workplace separately if you want to show the third row of tiles.

Additionally, there is the list of workplaces to the left. Here, you can select the workplaces for which details are displayed on the right-hand side.

## List of workplaces

The workplace list shows all workplaces/machines assigned to the terminal. If many workplaces are assigned, swipe to get to the workplaces displayed further down.

This information is shown for the workplaces:

### Machine/workplace number

Shows the machine and/or workplace number.

### Status

The status is displayed for each machine in color on the left-hand side and also the status text is colored. Coloring is as follows:

- green:           production
- yellow:         assigned status
- red:            not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

### Quantities

On the right-hand side, the first figure shows the produced yield in green and the red figure shows the produced scrap.



If you have enabled the *Compensate manual quantities* option (e.g. set off scrap against yield) and the machine list also shows shift-related quantities, they will not be updated immediately. The application only updates the quantities, once the lists have been reloaded.

### Unit for yield and scrap

If no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on, the primary quantity unit of the operation is displayed.

## Workplace tiles

The workplace tiles provide the following details:

### Workplace/machine

No workplace or machine number is displayed.

### Short name / group

The short name of the workplace or machine and the machine group are displayed.

**Status**

The status is displayed in color and as status text on the left-hand side. Coloring is as follows:

- green:           production
- yellow:         assigned status
- red:            not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

**Machine image**

Shows the picture of the machine stored in the configuration of workplaces.

**Clocks**

Shows the recorded machine cycles of the current shift.

**Start / Duration [hrs:min]**

Point in time since the status has been available and the resulting duration at the current point in time.

With BDE workplaces<sup>1</sup>, this point in time refers to the last manual status change. With MDE workplaces, it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was last changed manually or to the time of the last shift change.

**Target/actual cycle**

Current target and actual cycle of the workplace.

The largest target cycle of all operations logged on to the workplace is shown. The largest target cycle is transferred to the MDE for monitoring.

If the target cycle is smaller than the minimum cycle time, the target cycle is still shown.

If an operation is logged off or interrupted, the largest target cycle of the remaining operations is identified and displayed. After logoff or interruption of the last operation at the workplace, the last target cycle set is still displayed.

If no operation is logged on, the target cycle specified in the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that last applied.

**Yield / Scrap**

Yield and scrap quantities of the current shift produced at the machine/workplace.

---

<sup>1</sup> An MDE workplace is a workplace that is assigned to a terminal, which runs in the "MDE" operation mode. Otherwise, it is a BDE workplace.

## KPIs: OEE, utilization efficiency, scrap ratio



This function is only available if you enable the extension [aipkpi](#).

Shows the KPIs OEE, utilization efficiency and scrap ratio. The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached. If you have not defined limit values, the application shows the KPI in gray, otherwise in the color you defined for exceeding/not reaching limit values. For further information on the configuration, refer to the document [MDE KPI Configuration.pdf](#).



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

## Linked functions

If you click the workplace status, the dialog for changing the status opens. If you click one of the other tiles, the dialog opens where you can start the functions available for the selected workplace.



The buttons displayed depend on the selected workplace. The button *Lock production status*, for example, is only available for MDE machines.

## List of operations logged on

The middle area on the right-hand side shows the logged on operations as tiles. The following data is shown:

### MES order number

Order number and operation number of the operation logged on. The combination of these two numbers is the *MES order number*.

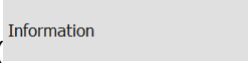
### Article

Article defined for the operation.

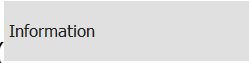
### Quantities (target / yield / scrap)

Shows the target quantity defined for the operation, the produced yield and the scrap. The yield and scrap quantities integrate the counter readings of the available machine connections.



This icon is displayed if at least one note has been recorded for this operation in the graphic planning board of the client that must be shown on the terminal. To display the note(s), click the icon or click the operation and select the *Information* button (  ).



This icon is displayed if at least one long text is stored for this operation. Click this icon to display the long texts or click the operation and select the button *Information* (  ).

If you click an operation, a screen opens that shows the workplace/operation data. Via this screen, you can also select the operation-related functions.

## Operation tiles

In addition to the fields already described, the operation tiles also show the following data:

### Comments

This tile shows the user fields 53 and 54 (alphanumeric, 20 characters) of the operation. To edit these fields, you must store a respective user field key for the operation, which includes these two fields.

### Completion in %

The bar shows the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

### Since logon (target / yield / deviation)

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time when no production lock has been set for the machine). If the terminal program has been restarted after the OP logon, no value can be calculated.

Calculation:

Target Since Logon = Net Running Time[sec] \* Partitioning/Target Cycle[sec/stroke]

Net Running Time: Time since logon while the production lock has not been set. This calculation does not integrate the breaks specified in the shift model or the status times posted to RPA 12 (resource performance account).

Deviation (in percent) between the calculated target quantity since logon and the quantity which has actually been produced “since logon”.

Calculation: Deviation[%] = 100% \* (Yield Since Logon - Target Quantity Since Logon) / Target Quantity Since Logon.

As of CTAIP 8.2.1.32:

Up to now, the target quantity since OP logon was only calculated if the workplace/machine was assigned to a terminal with operation mode "MDE processing". As of CTAIP 8.2.1.32, the target quantity since OP logon is also calculated if the terminal is configured with operation mode "BDE processing".

If the terminal has been restarted after an operation logon, the target quantity cannot be calculated correctly. To improve transparency, an "\*" (asterisk) is shown behind the target quantity (since OP logon) in this case. The asterisk indicates that the target quantity now displayed no longer refers to the time of the operation logon, but to the time of the terminal restart.

While the workplace/machine status 999 is displayed, the target quantity since OP logon is "---". If the status 999 is again changed within a "free shift", the target quantity since OP logon is calculated using the point in time of the OP logon, of the shift start or of the terminal start.

To disable the calculation of the target quantity since OP logon on the terminal, you can use the configuration CalcTargetYieldSinceLogon=0 in the hytnrcfg.ini. The quantity is then displayed using "---".



With workplaces/machines that are configured as "Machining centers", the calculation of the target quantity since OP logon is generally disabled because this calculation contradicts the principles of the machining center.

### Planned duration

The field *Planned duration* displays the target processing time of the operation in format [h:mm].

### Partitioning

Calculate the displayed partitioning as follows:

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| TLG <sub>M</sub>  | Partitioning of the workplace/machine (TLG in mnr.lst) |
| DIV <sub>M</sub>  | Pulse factor of workplace/machine (IMPFAKT in mnr.lst) |
| TLG <sub>AG</sub> | Partitioning of the operation (TLG in anr.lst)         |

The application shows the calculated partitioning without decimal places, provided it is an integer value.

In case the partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

### Displaying the "3rd list"

The third list is optional. You can configure the third list in the configuration of workplaces. The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)

- List of resources logged on to the currently selected workplace (WRM)
- List of materials/input batches (MPL/TRT) logged on to the currently selected workplace
- List of output batches produced in the currently selected operation (MPL/TRT)

In case you have enabled several lists, you can switch between these lists in the header line that is located above the third list. Activated lists can be selected one after the other.

### Maintenance status

If you have purchased the license for the maintenance calendar, the maintenance status is displayed using a yellow or a red field showing a wrench. The color displayed depends on the required maintenance activity.

### Calling functions

The functions available are assigned to the relevant objects. Example: The functions *Log person off* and *Log all staff off* are displayed if you click on a person logged on.

## 1.3 Icon view of workplaces

You can enable this view in the configuration of terminals via the client. Then open this view by clicking the button "< Overview" in the main view. This view shows workplaces in a clear structure and with an image. It shows important information on the single workstations:

|                            |                            |                              |                              |                            |                            |
|----------------------------|----------------------------|------------------------------|------------------------------|----------------------------|----------------------------|
| Workplace/machine<br>33021 | Workplace/machine<br>40312 | Workplace/machine<br>406457A | Workplace/machine<br>406457B | Workplace/machine<br>79667 | Workplace/machine<br>79668 |
| Status<br>PRODUCTION       | Status<br>OUT OF MATERIAL  | Status<br>PRODUCTION         | Status<br>PRODUCTION         | Status<br>PRODUCTION       | Status<br>PRODUCTION       |
| Yield<br>833               | Yield<br>1381              | Yield<br>2500                | Yield<br>66                  | Yield<br>51                | Yield<br>78                |
| Scrap<br>4                 | Scrap<br>5                 | Scrap<br>13                  | Scrap<br>0                   | Scrap<br>1                 | Scrap<br>0                 |
|                            |                            |                              |                              |                            |                            |

Please note: The actual display can be different to the above illustration.

A colored bar to the left of the image indicates the current status of workplaces/machines:

- green: production
- yellow: assigned status
- red: not assigned

Each tile includes the workplace/machine number, the status (text) of the workplace, the yield and scrap quantity and the image of the workplace.

A colored background with a caliper and/or wrench to the right of the image indicates if an inspection or maintenance is due for the workstation.



If you enable the extension [aipkpi](#), the application also shows the KPIs OEE, utilization efficiency and scrap ratio.

The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

If you click on a tile, the previously described main view is displayed and the workplace is automatically selected. From there, you can perform the postings for the selected workplace.

Use the option "< Overview" to exit the main view and to return to the icon view.



As part of the advanced configuration options, you can customize the layout of display lists, the displayed data fields and functions. For technical reasons, however, you cannot change the sort sequence of display lists in the main view of the terminal.

As of AIP 8.2.2.28, you can automatically change from the main view with tiles to the icon view of workplaces after a configured time. The configuration `AUTOMATIC-CHANGE-TO-START-DISPLAY` is described in the document [AIP2\\_Configuration\\_hytnrcfg.pdf](#).